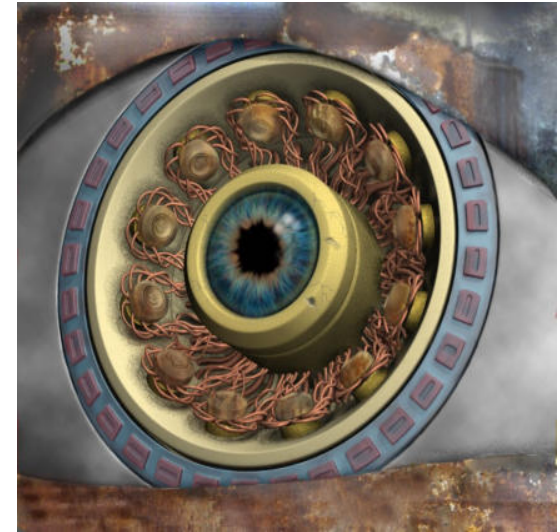
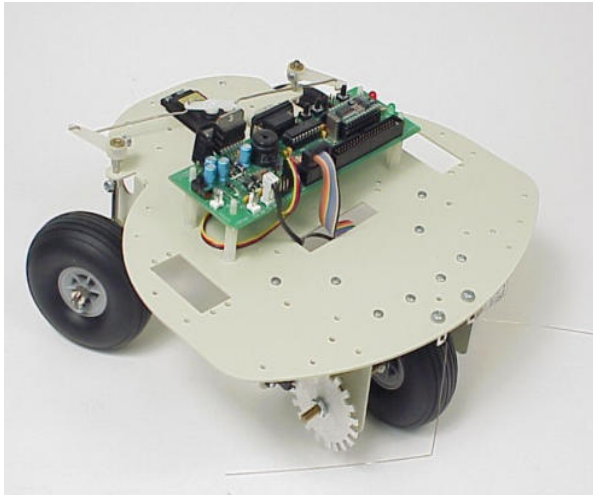




Introducción a la visión para robots

Robótica y Percepción Computacional

4 Curso. Graduado en Ingeniería Informática

Luis Baumela
Departamento de Inteligencia Artificial
Universidad Politécnica de Madrid



POLITÉCNICA

Robótica y Percepción Computacional

Temario:

1. Introducción
 - 1.1 Introducción a la robótica
2. Navegación
 - 2.1 Control
- 3. Percepción**
 - 3.1 Segmentación de imágenes digitales**
 - 3.2 Reconocimiento de marcas
4. Integración
 - 4.1 Pruebas de integración en el robot

Why computer vision in a robotics course?

What is a robot?

A machine that can move, plan, act and sense.

What are typical senses in living creatures?

Are there robotics counterparts?

- taste
- smell
- hearing
- touch
- sight

Why computer vision in a robotics course?

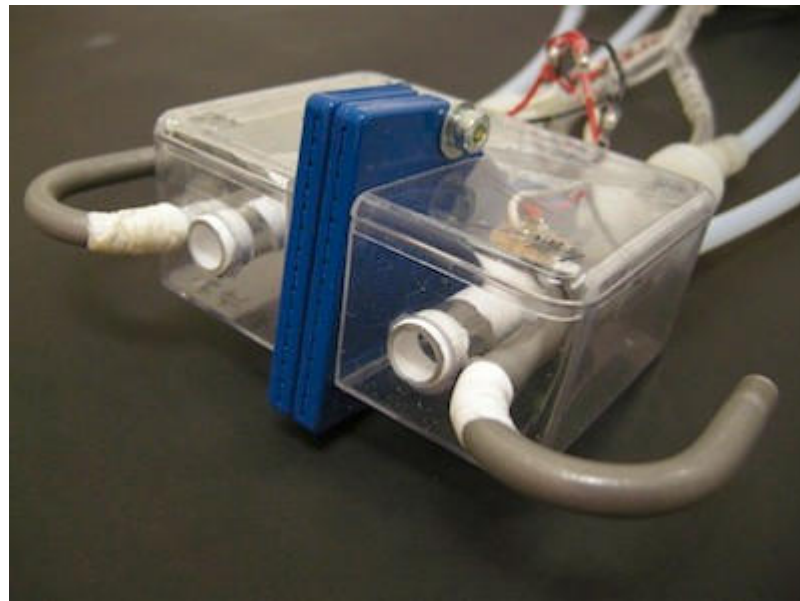
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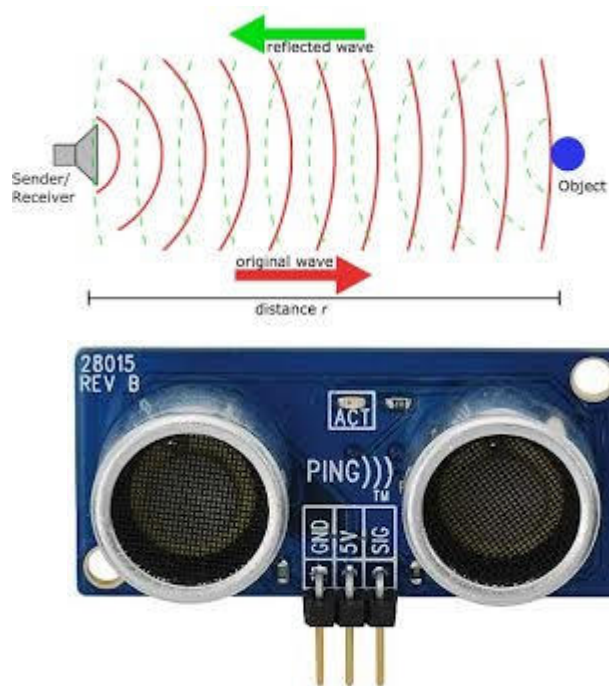
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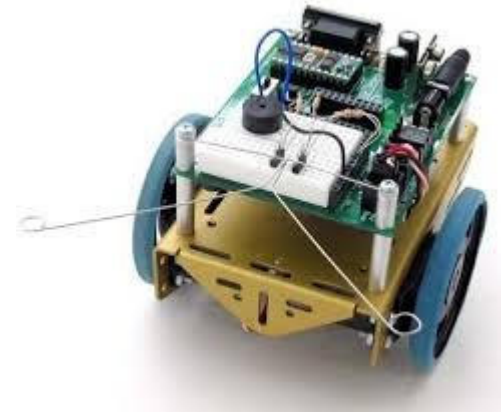
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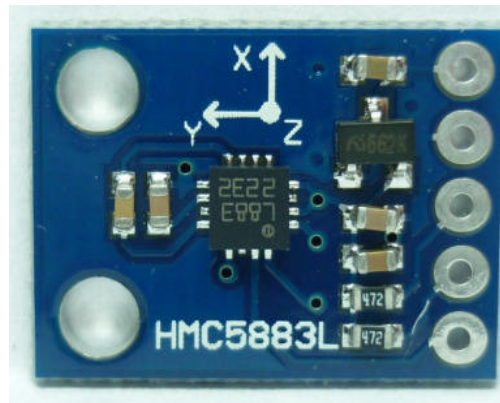
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- smell
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- touch
- sight
- **magnetism**

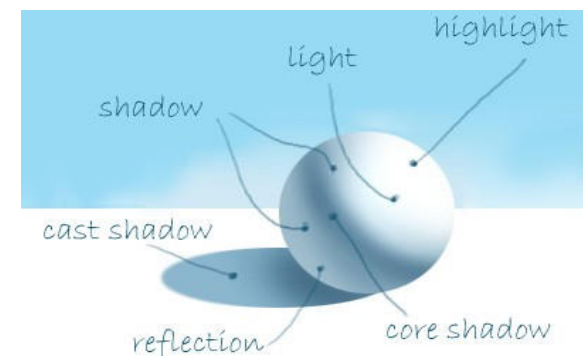
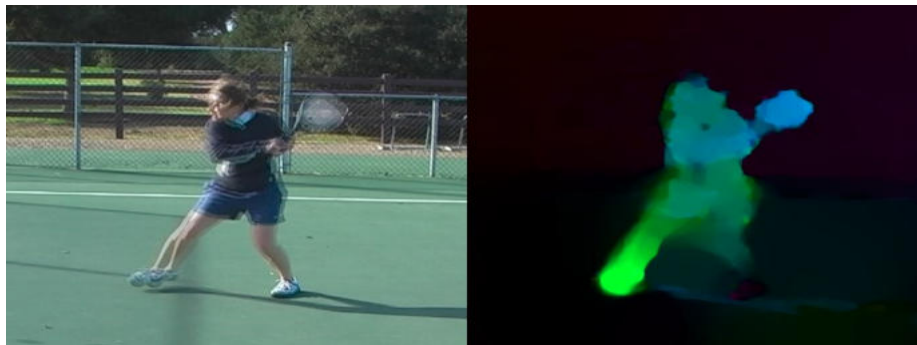
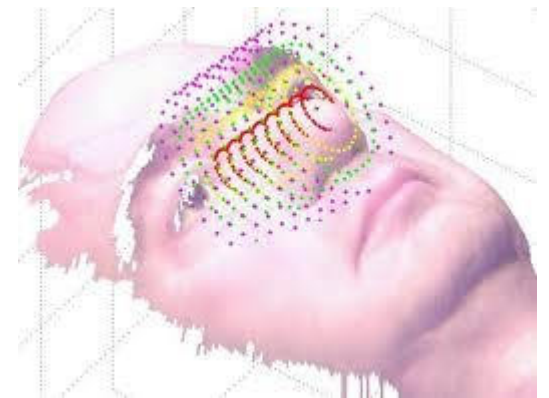
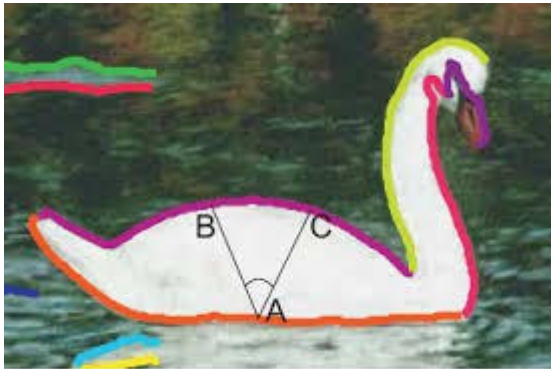


Why computer vision in a robotics course?

Why vision?

Vision provides an enormous amount of information.

- shape, colour, texture,
- illumination, motion, depth



Why computer vision in a robotics course?

Why vision?

Vision provides an enormous amount of information.

- shape, colour, texture,
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It is an inexpensive sensor



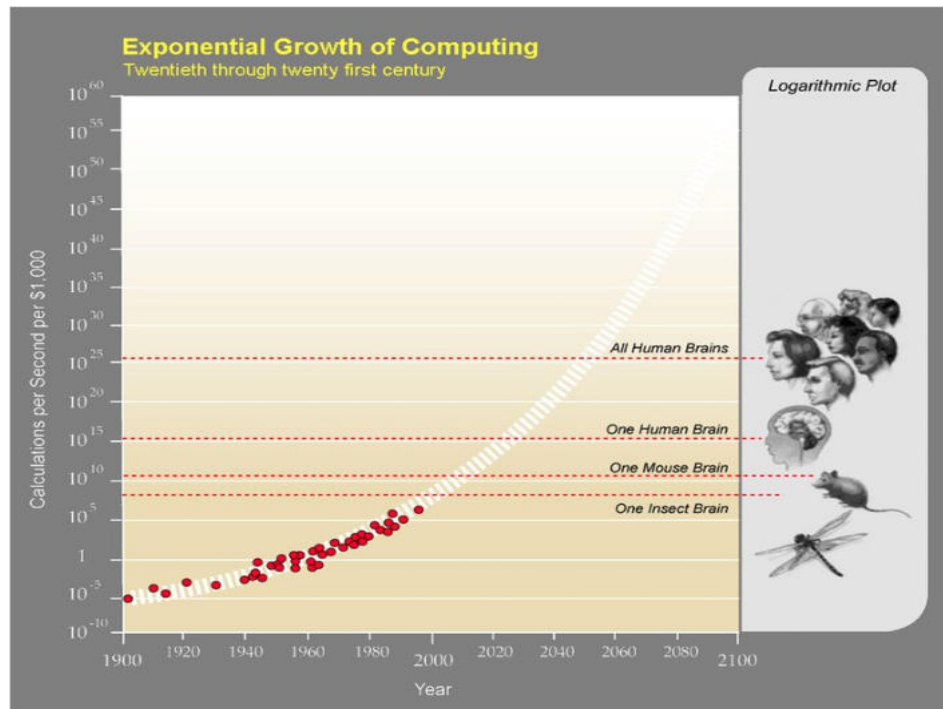
Why computer vision in a robotics course?

Why vision?

Vision provides an enormous amount of information.

It is an inexpensive sensor.

Computing power grows exponentially.



Computer Vision

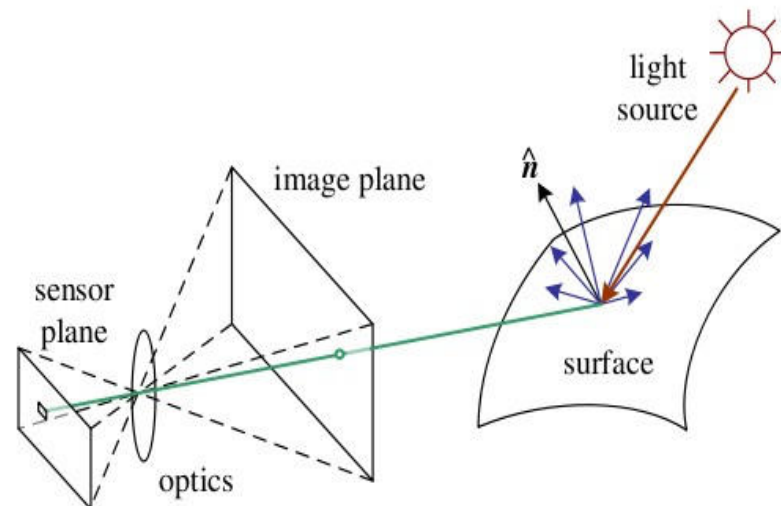
Vision is the process of discovering from images what is in the world and where it is (D. Marr)

What is the goal of computer vision?

Make machines that see; i.e. obtain information from images.

Challenge: Vision is an *inverse* problem

45	60	98	127	132	133	137	133
46	65	98	123	126	128	131	133
47	65	96	115	119	123	135	137
47	63	91	107	113	122	138	134
50	59	80	97	110	123	133	134
49	53	68	83	97	113	128	133
50	50	58	70	84	102	116	126
50	50	52	58	69	86	101	120



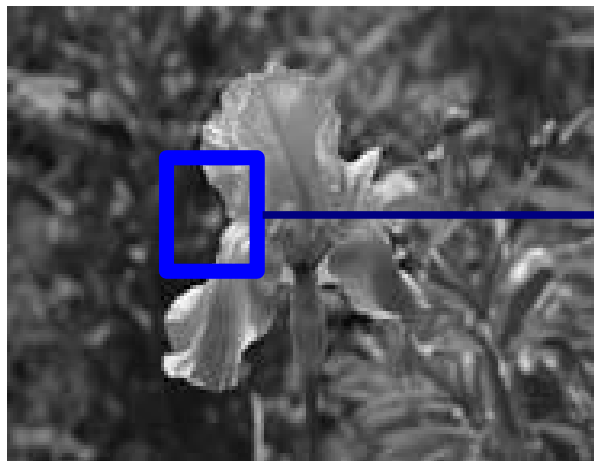
Computer Vision

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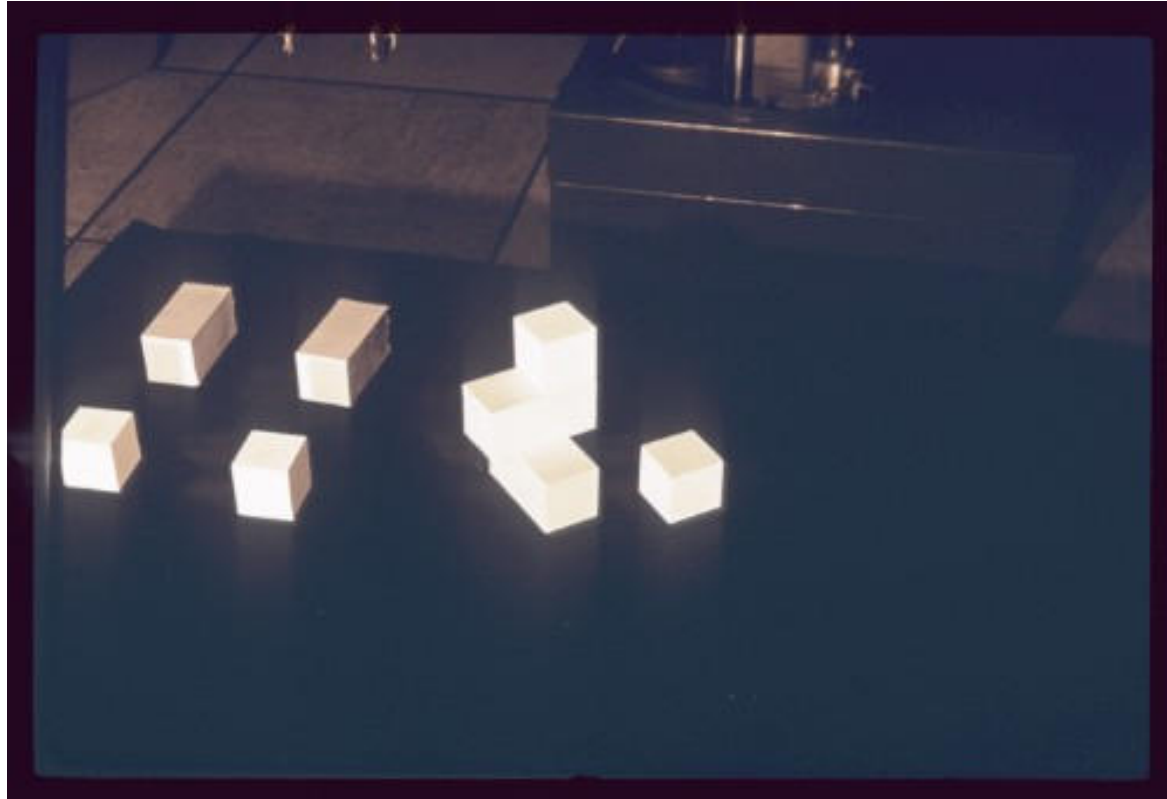


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49	53	68	83	97	113	128	133
50	50	58	70	84	102	116	126
50	50	52	58	69	86	101	120

Computer Vision

MIT Vision Project, 1966.
Solve image interpretation in a summer!

MIT AI Lab Copy Demo project.

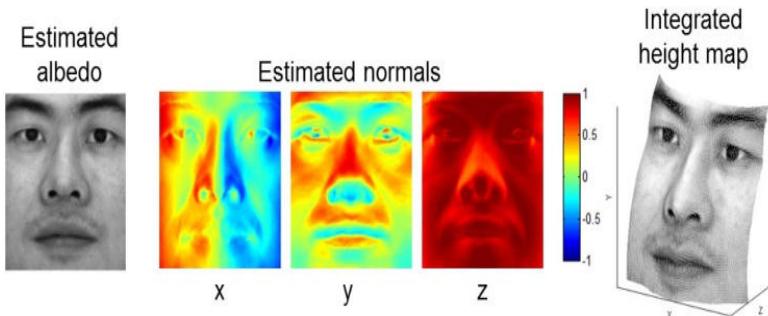
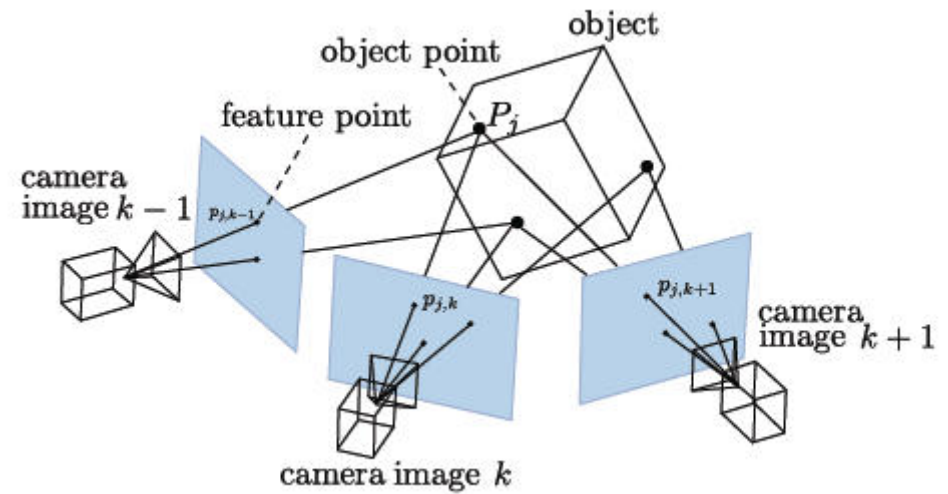


Computer Vision

Vision is an *inverse* problem

- **Reconstruccion**

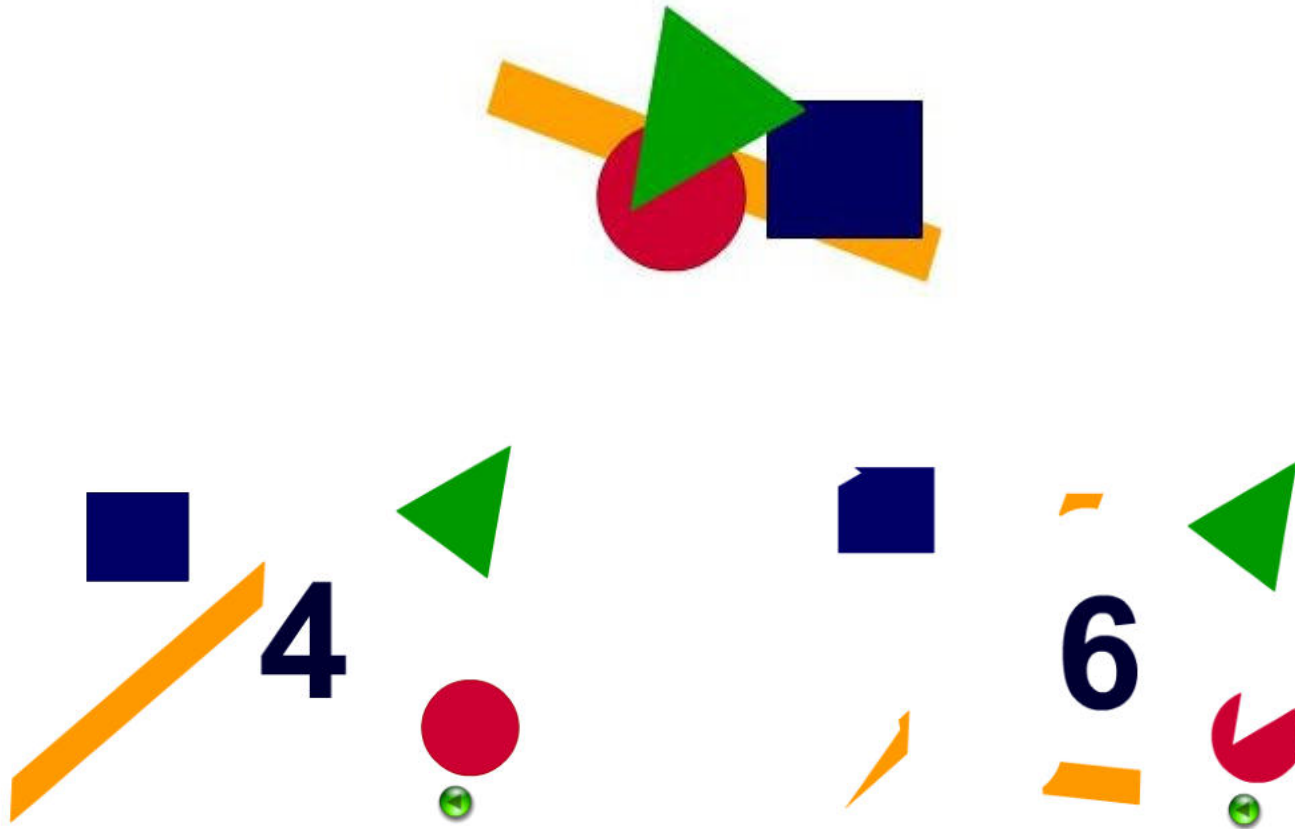
- Shape from X (estéreo)
- Shadows and shading
- Shape from defocus



Computer Vision

Ambiguities and optical illusions

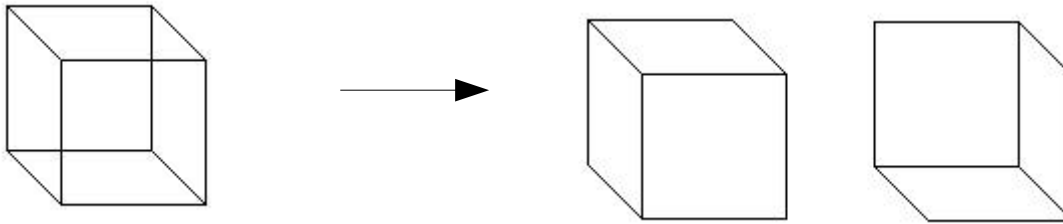
How many objects are there?



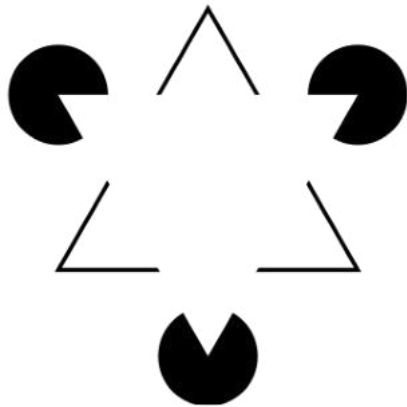
Computer Vision

Ambiguities and optical illusions

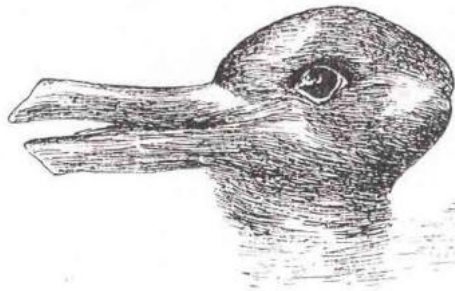
Which side of the cube is in front?



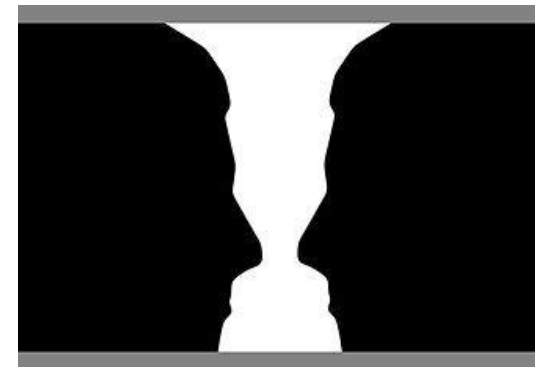
What object can you see?



Kanizsa's Triangle



Rabbit-duck illusion

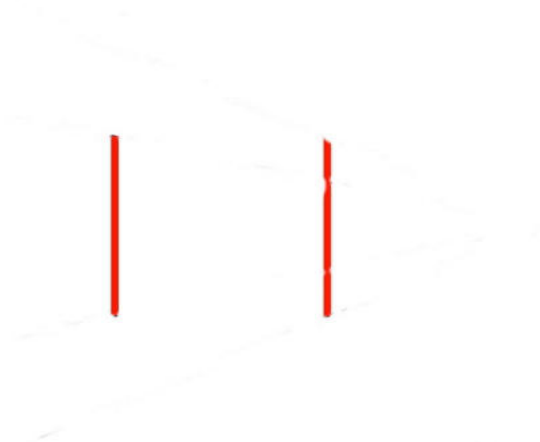
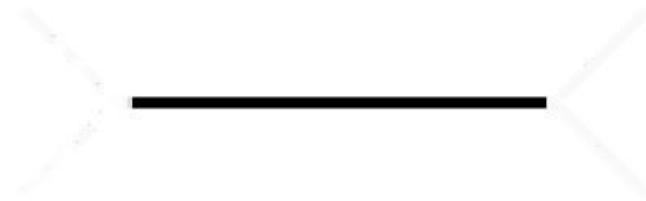
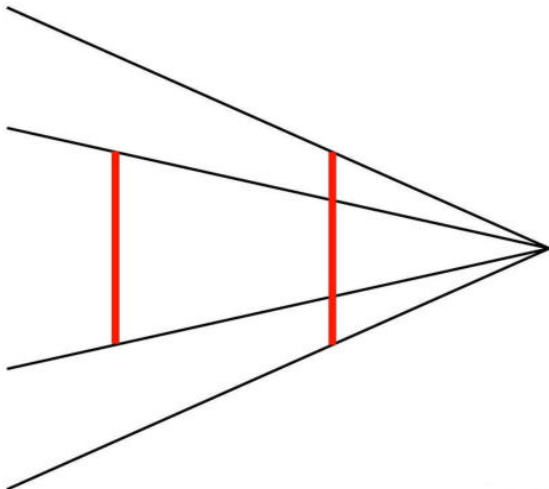
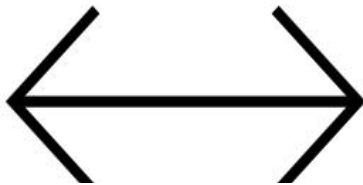
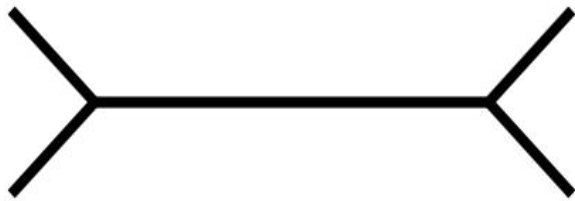


Reversible figures and vase

Computer Vision

Ambiguities and optical illusions

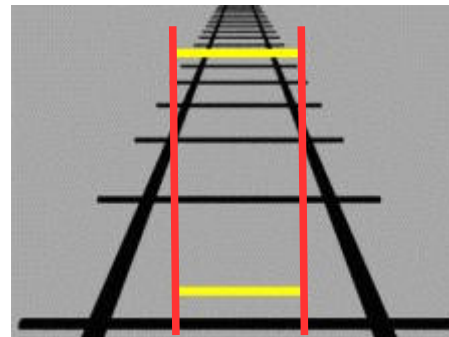
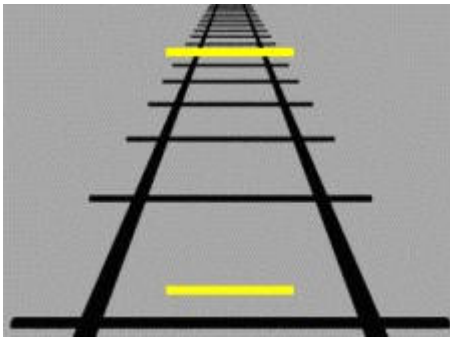
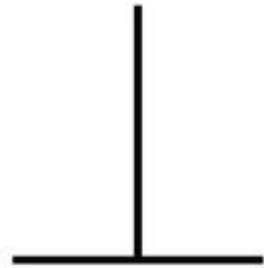
In the perceived size of objects.



Computer Vision

Ambiguities and optical illusions

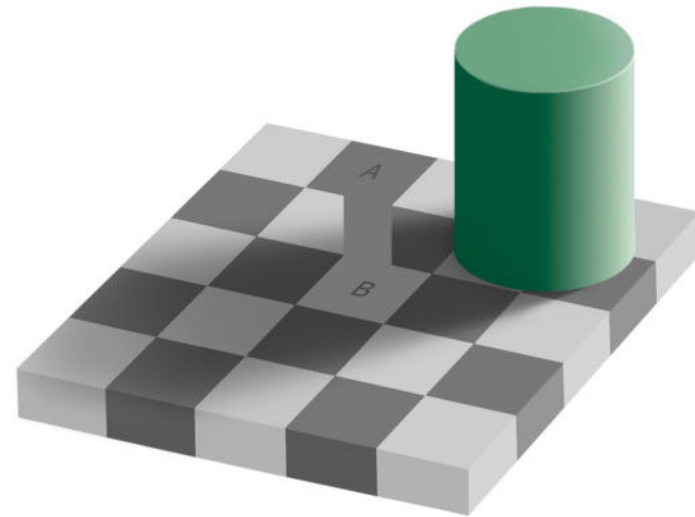
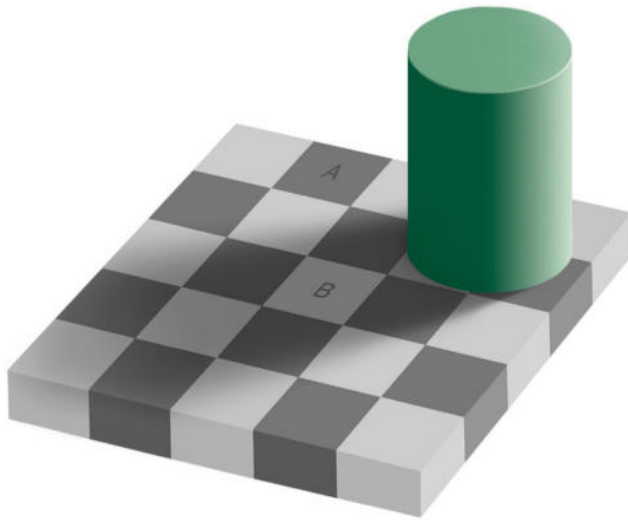
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Computer Vision

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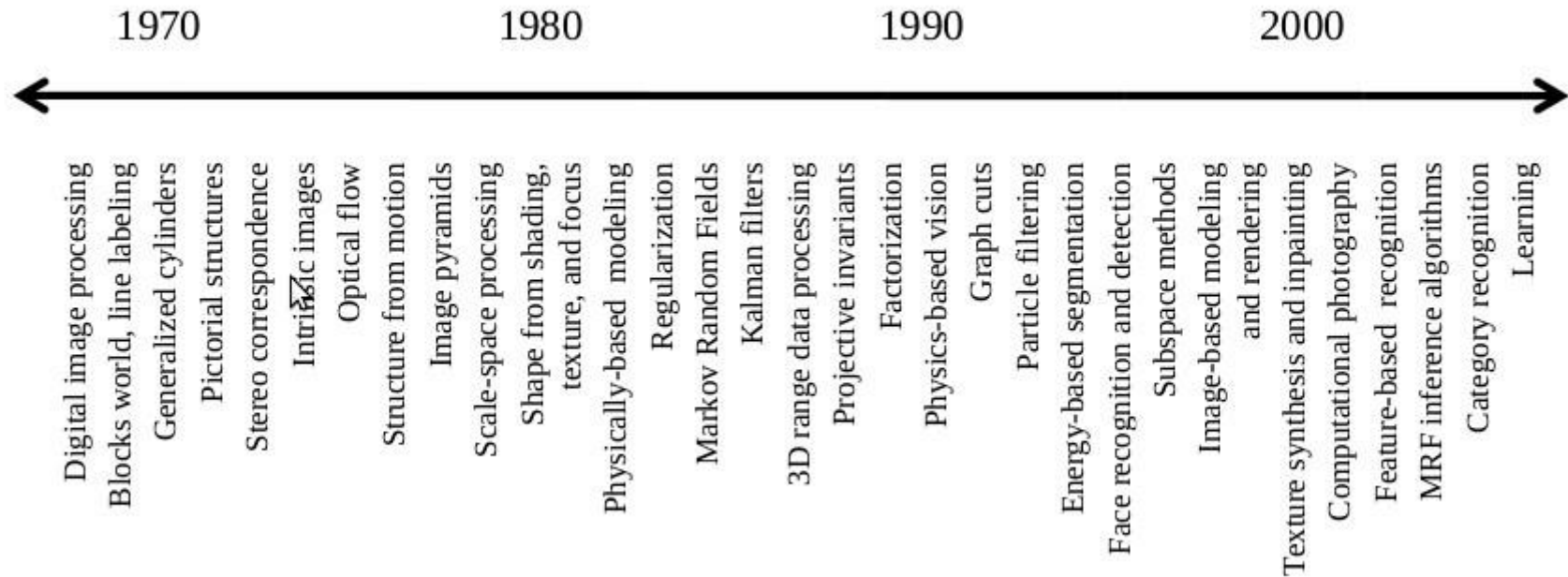
Colour and brightness.



Computer Vision

History of computer vision.

- 1956 Darmouth conference origin of AI.



Robotics in a computer vision course?

Does computer vision benefit from robotics?

Vision is an active process

Robotics in a computer vision course?

Does computer vision benefit from robotics?

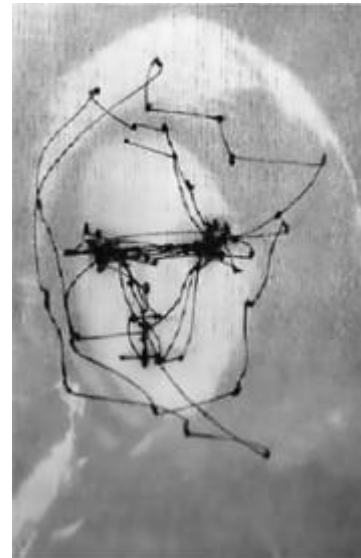
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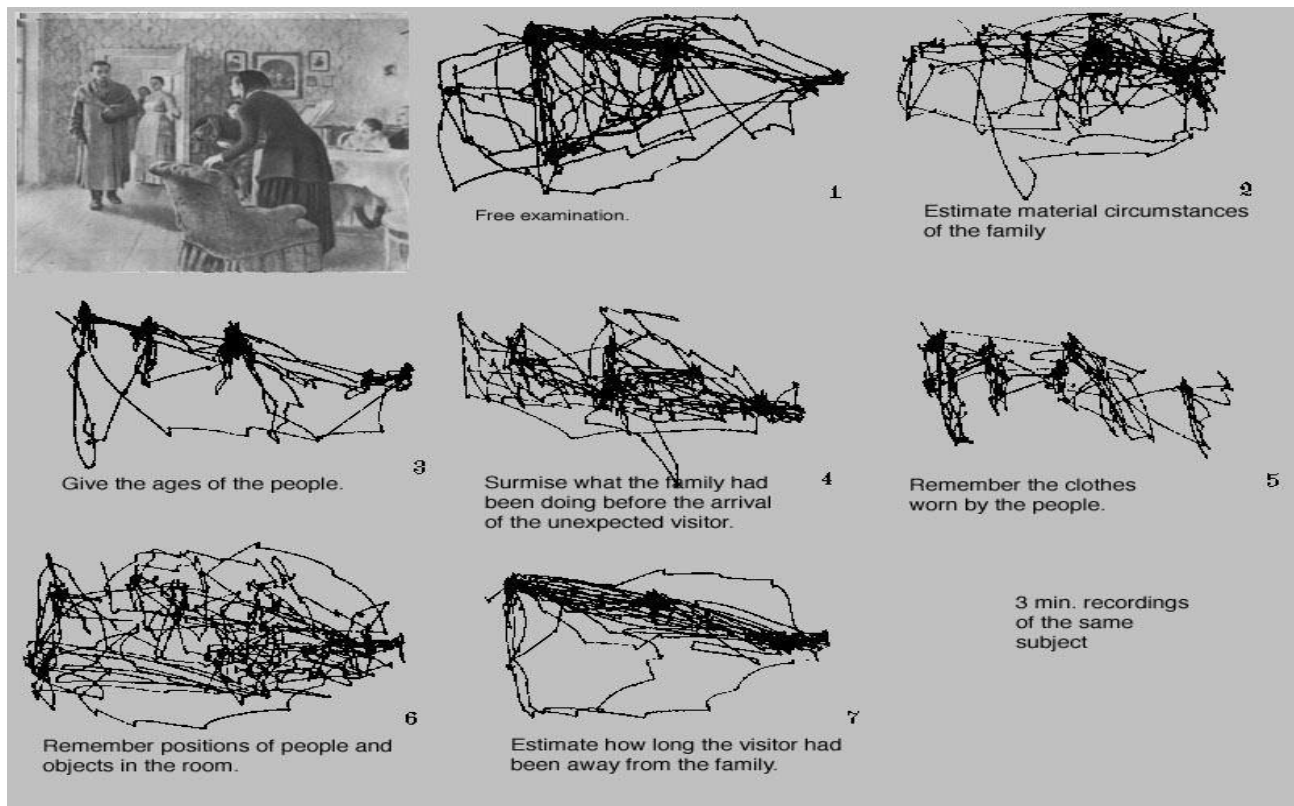


Yarbus, A. L. (1967), Eye Movements and Vision, New York: Plenum

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Yarbus, A. L. (1967), Eye Movements and Vision, New York: Plenum

Computer vision applications

- **Automated cartography**
- **Autonomous driving**
- **Autonomous navigation (vision in Mars)**
- **Visual interaction (3D cameras)**

Computer vision applications

Automated cartography



<http://www.sensefly.com>

<http://pix4d.com>

Computer vision applications

Autonomous driving

- **70s.** Stanford Cart, Sakey



Computer vision applications

Autonomous driving

- **70s.** Stanford Cart, Sakey
- **80-90s.** Dickmanns cars



- 1987. 96 km/h sin tráfico
- 1994 con tráfico en París

Ernst Dickmanns' First Mercedes-Benz Prototype, the VaMoRs

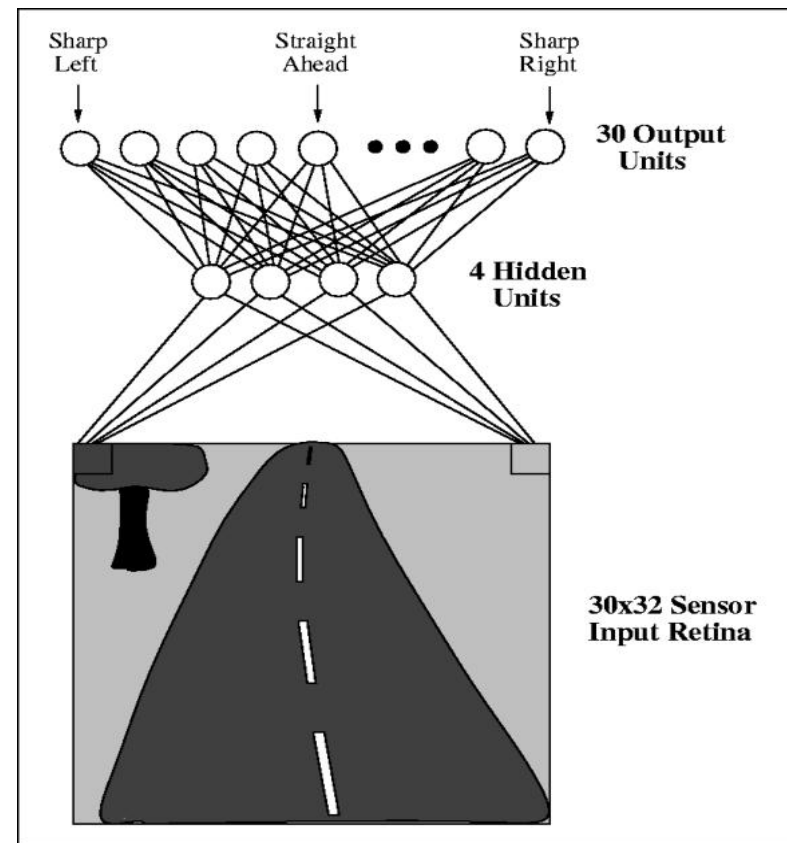
Computer vision applications

Autonomous driving

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- **80-90s.** Dickmanns cars
- **1995.** Kanade Pomerlau ALVINN



Sólo controla dirección



Computer vision applications

Autonomous driving

- **70s.** Stanford Cart, Sakey
- **80-90s.** Dickmanns cars
- **1994.** Kanade-Pomerlau ALVINN
- **2005.** DARPA's no hand across america



- no traffic
- dirt roads
- GPS: course predefined by 2935 GPS points (up to 4 GPS points per street curve).
- Obstacle types known in advance

Computer vision applications

Autonomous driving

- **70s.** Stanford Cart, Sakey
- **80-90s.** Dickmanns cars
- **1994.** Kanade-Pomerlau ALVINN
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- **2007. DARPA**s urban challenge



Computer vision applications

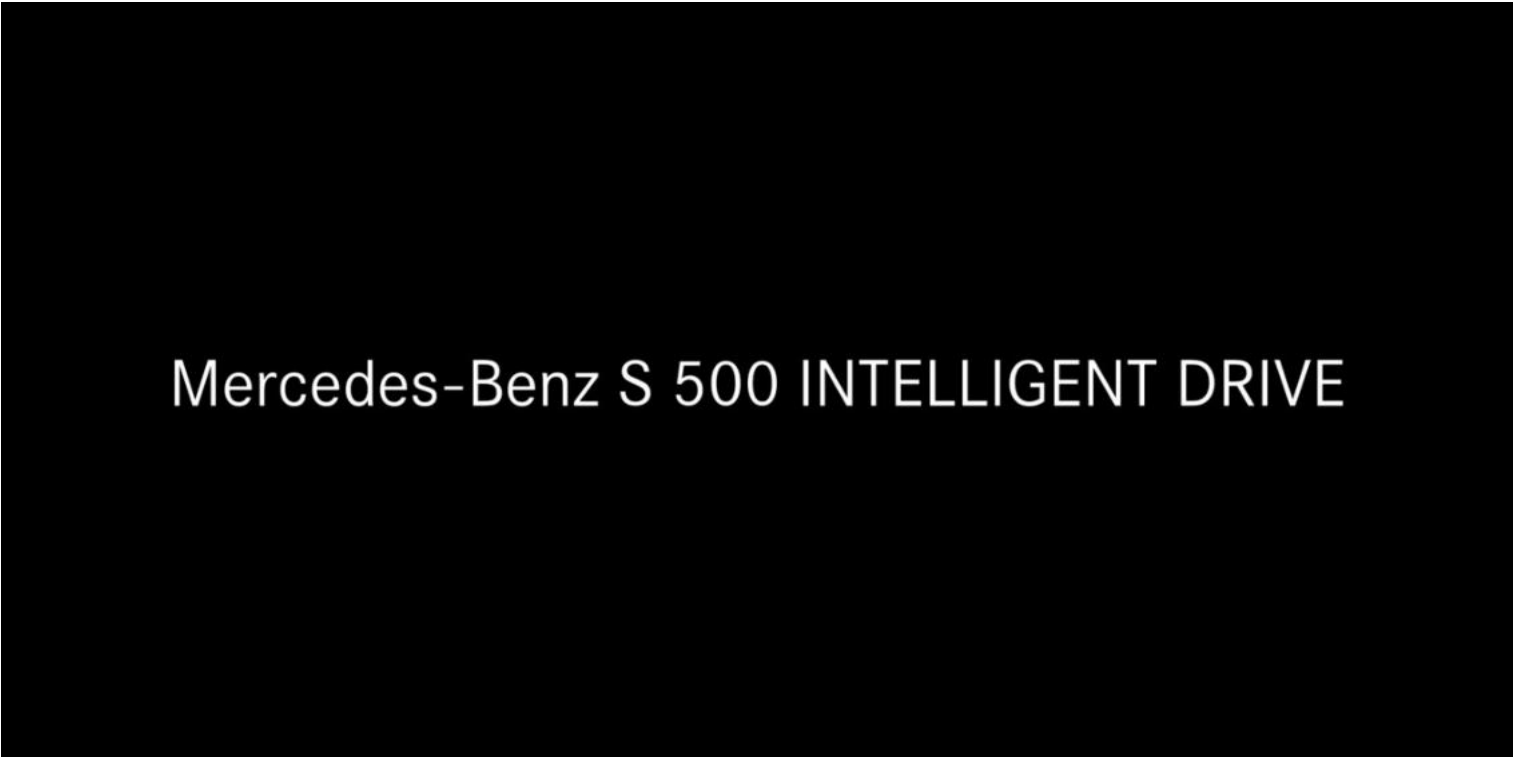
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- **80-90s.** Dickmanns cars
- **1994.** Kanade-Pomerlau ALVINN
- **2005. DARPA**s no hand across america
- **2007. DARPA**s urban challenge
- **2011.** Google self driving cars



Computer vision applications

Autonomous driving



Mercedes-Benz S 500 INTELLIGENT DRIVE

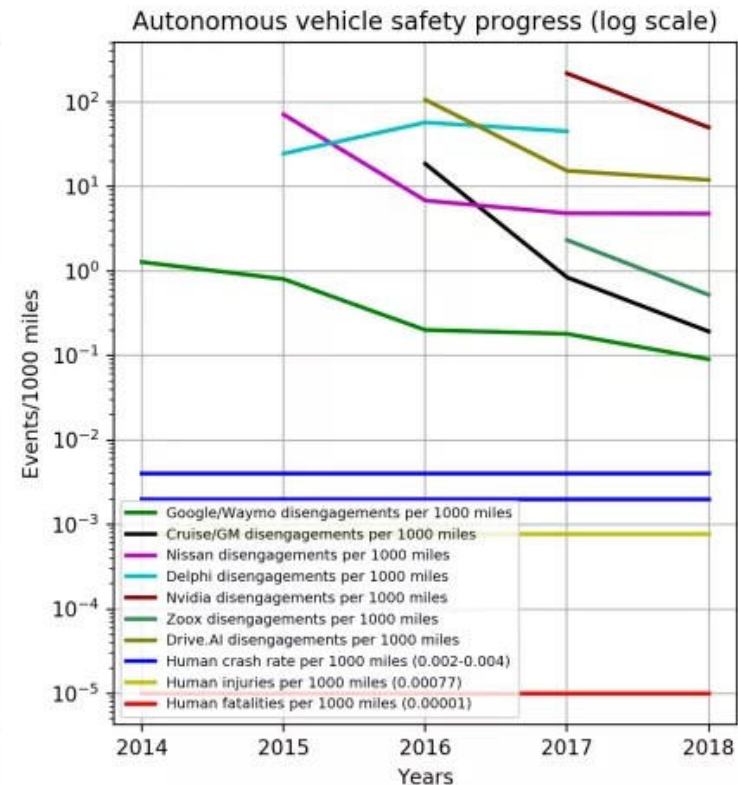
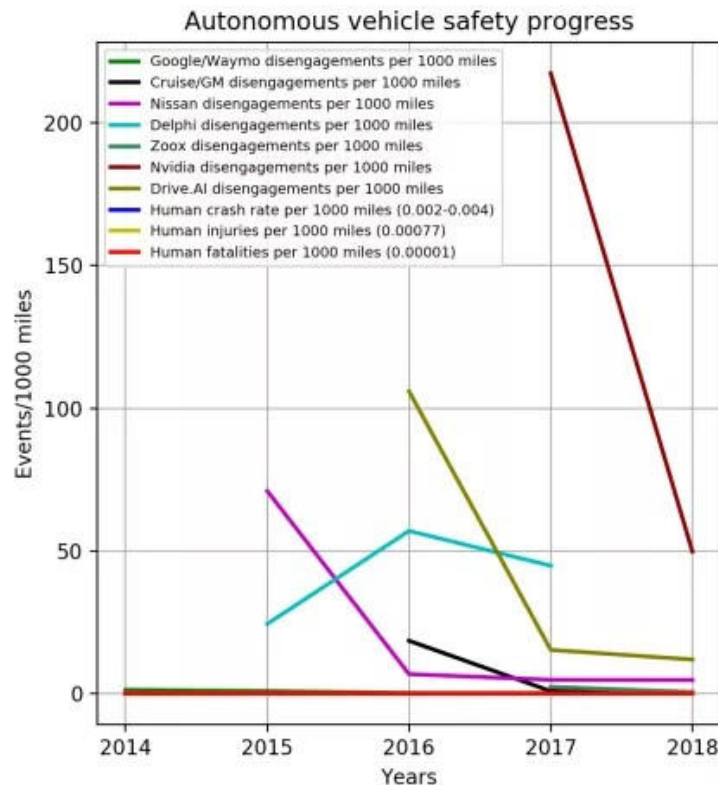
Mercedes Benz S500 Intelligent Drive

Computer vision applications

Autonomous driving

- California DMV disengagement data

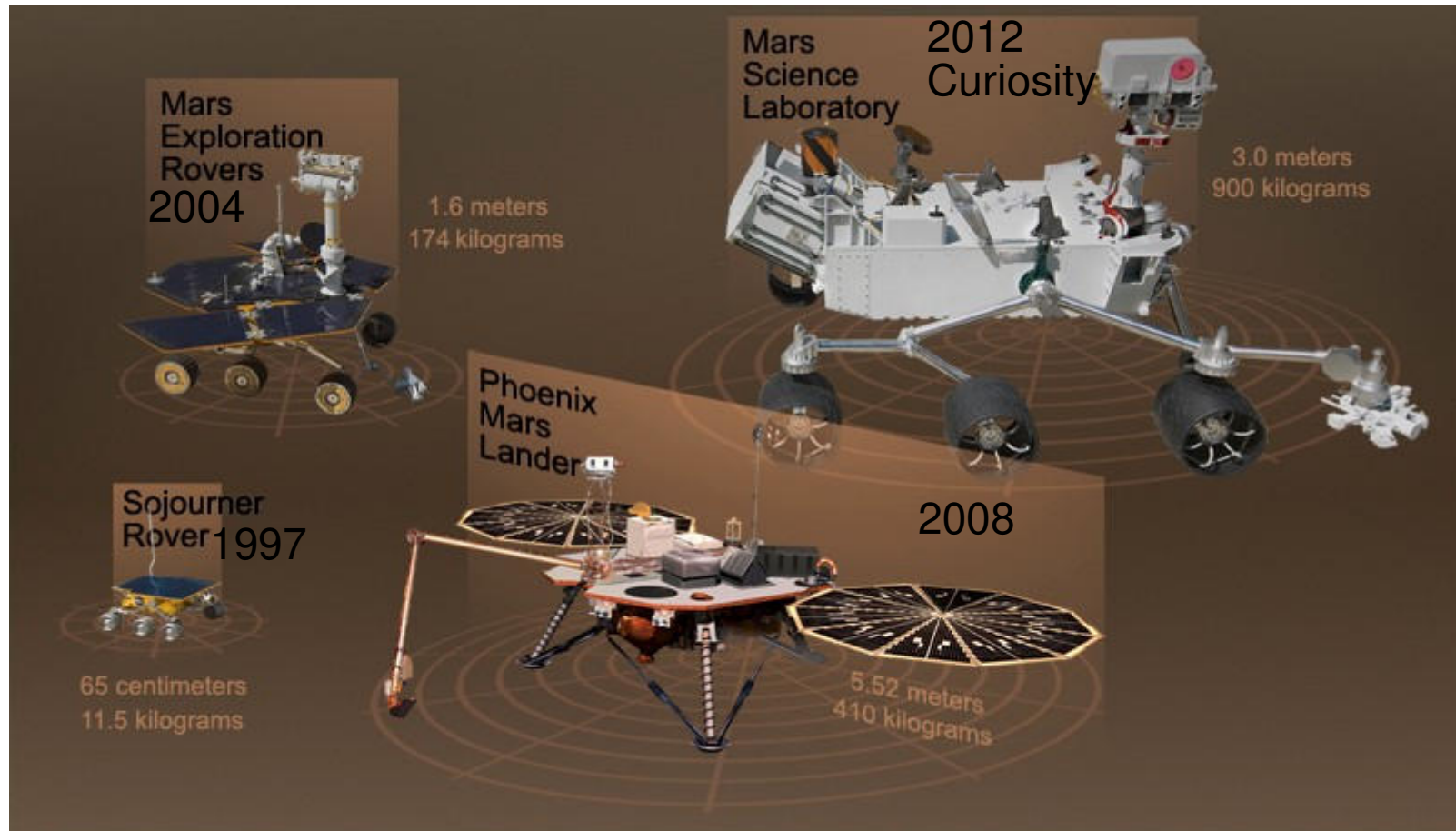
A disengagement is a situation in which a machine cannot be trusted and the human operator takes over to avoid any danger.



Computer vision applications

Autonomous navigation

Mars Rovers



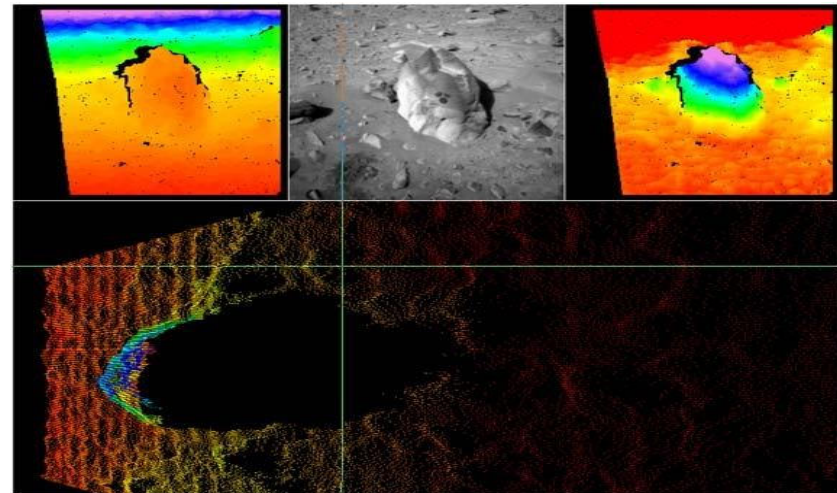
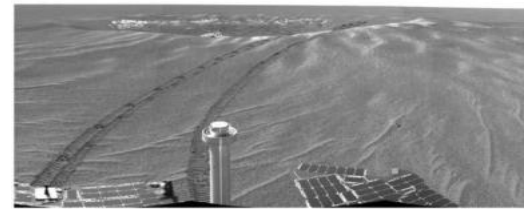
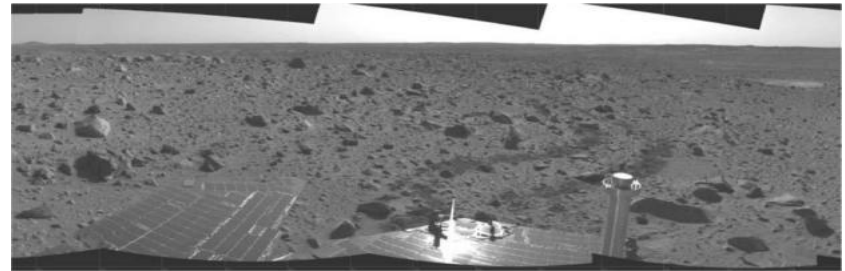
Computer vision applications

Autonomous navigation

Mars Rovers



Opportunity & Spirit
landed 2004
Opportunity – ended 2018
Spirit – ended 2011

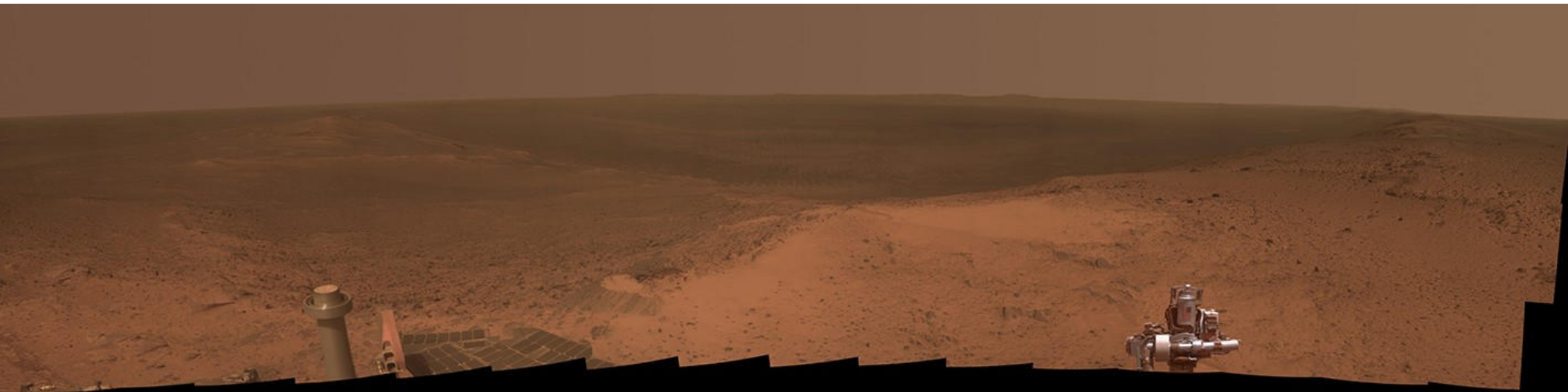


Matthies, et al. Computer Vision on Mars, Intl. J.. of Computer Vision, 2007

Computer vision applications

Autonomous navigation

Mars Rovers



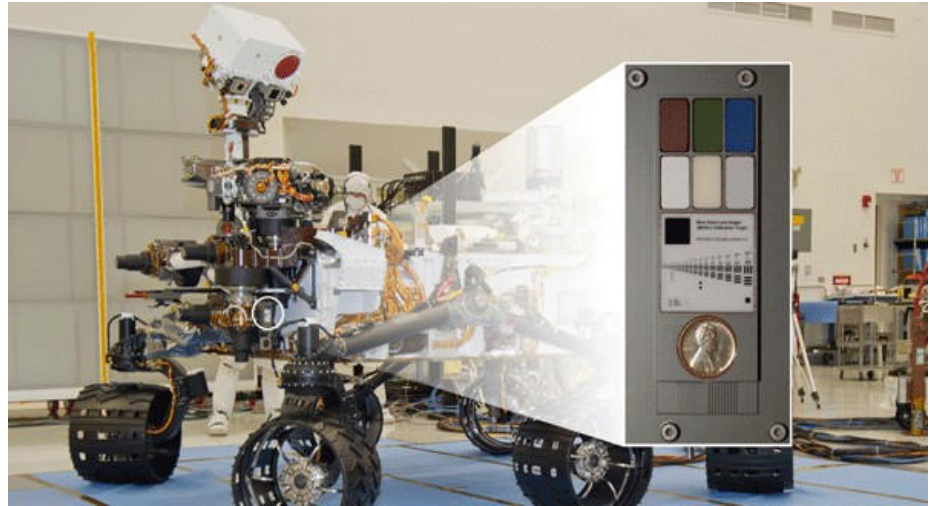
View of NASA's Mars Exploration Rover Opportunity from the top of the "Cape Tribulation" segment of the rim of Endeavour Crater. The rover reached this point three weeks before the 11th anniversary of its January 2004 landing on Mars.



Computer vision applications

Autonomous navigation

Mars Rovers



Calibration Target on Curiosity for Camera on Rover's Arm



Curiosity
Landed 2012
still in operation

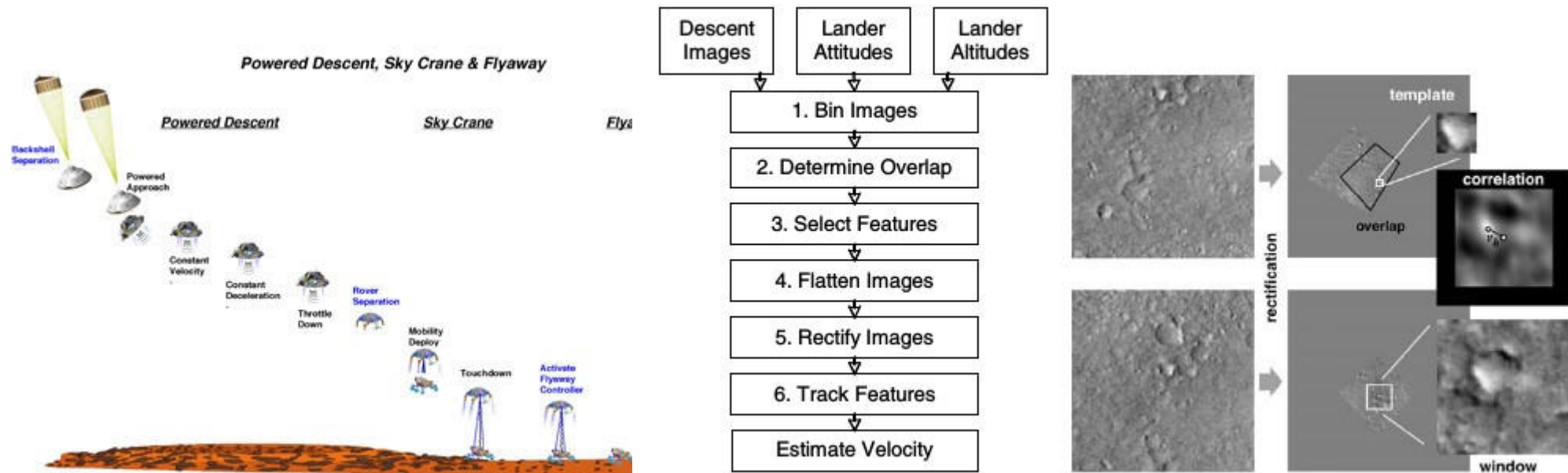


First image from a mars
rover choosing a target
using AEGIS.

Computer vision applications

Autonomous navigation

Landing ground speed computation using computer vision



Computer vision applications

Autonomous navigation

Industrial robots



Instalaciones de la fábrica de Seat en Martorell (Barcelona), septiembre 2019

Computer vision applications

- 3D Reconstruction/Tracking

Kinect

Microsoft, USA

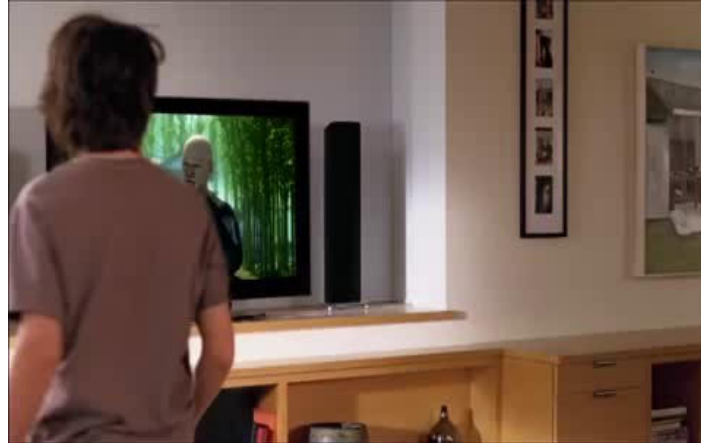


Computer vision applications

- 3D Reconstruction/Tracking

Kinect

Microsoft, USA

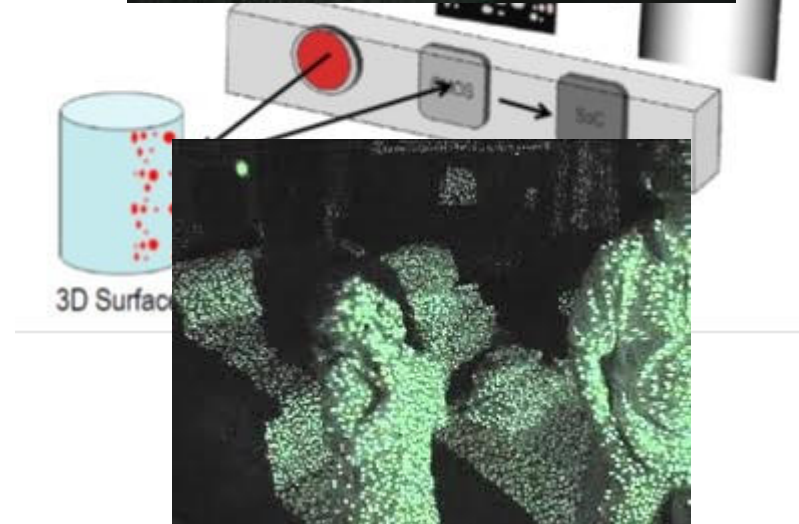
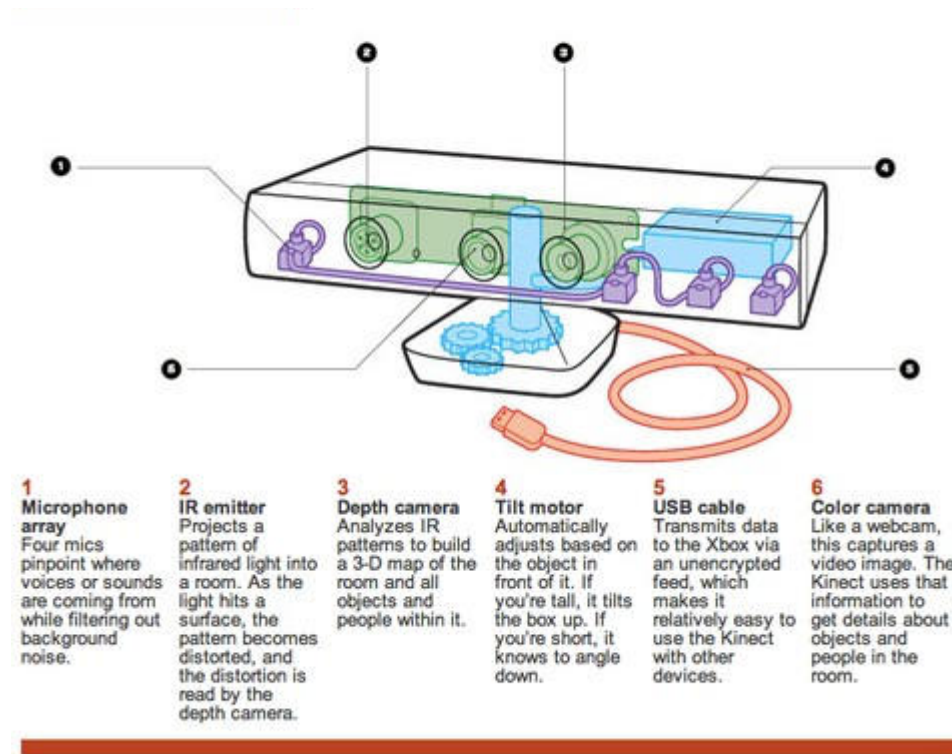


Computer vision applications

- 3D Reconstruction/Tracking

Kinect v1

Microsoft, USA

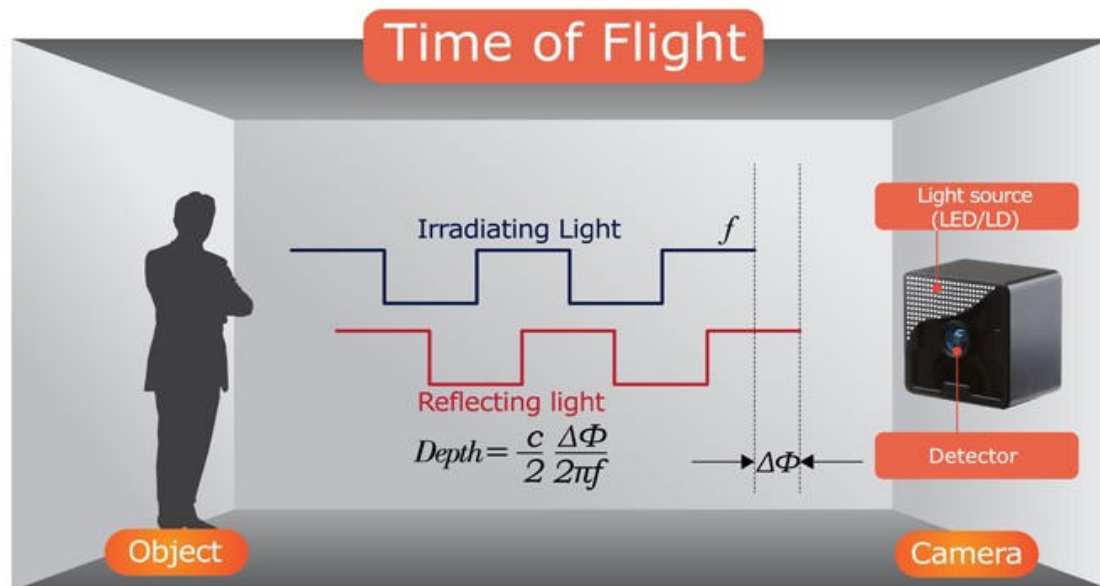


Computer vision applications

- 3D Reconstruction/Tracking

Kinect v2

Microsoft, USA



Computer vision applications

- 3D Reconstruction/Tracking

Kinect v1 vs. v2

Microsoft, USA



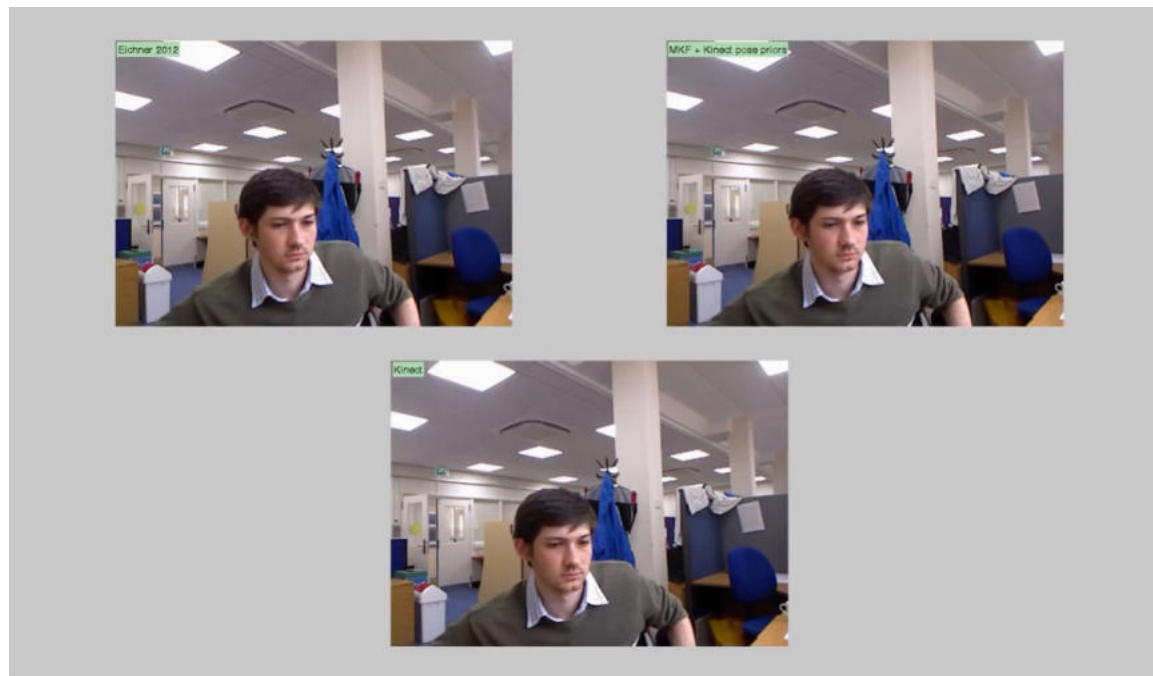
Feature	Kinect for Windows 1	Kinect for Windows 2
Color Camera	640 x 480 @30 fps	1920 x 1080 @30 fps
Depth Camera	320 x 240	512 x 424
Max Depth Distance	~4.5 M	~4.5 M
Min Depth Distance	40 cm in near mode	50 cm
Horizontal Field of View	57 degrees	70 degrees
Vertical Field of View	43 degrees	60 degrees
Tilt Motor	yes	no
Skeleton Joints Defined	20 joints	26 joints
Full Skeletons Tracked	2	6
USB Standard	2.0	3.0
Supported OS	Win 7, Win 8	Win 8
Price	\$299	TBD

Computer vision applications

- 3D Reconstruction/Tracking

Kinect

Microsoft, USA



Computer vision applications

- 3D Reconstruction/Tracking

Kinect

Microsoft, USA



Accurate, Robust, and Flexible Real-time Hand Tracking

Toby Sharp, Cem Keskin, Duncan Robertson, Jon Taylor, Jamie Shotton,
Ido Leichter, Alon Vinnikov, David Kim, Christoph Rhemann,
Yichen Wei, Daniel Freedman, Eyal Krupka,
Andrew Fitzgibbon, Shahram Izadi

Microsoft Research

Contact:

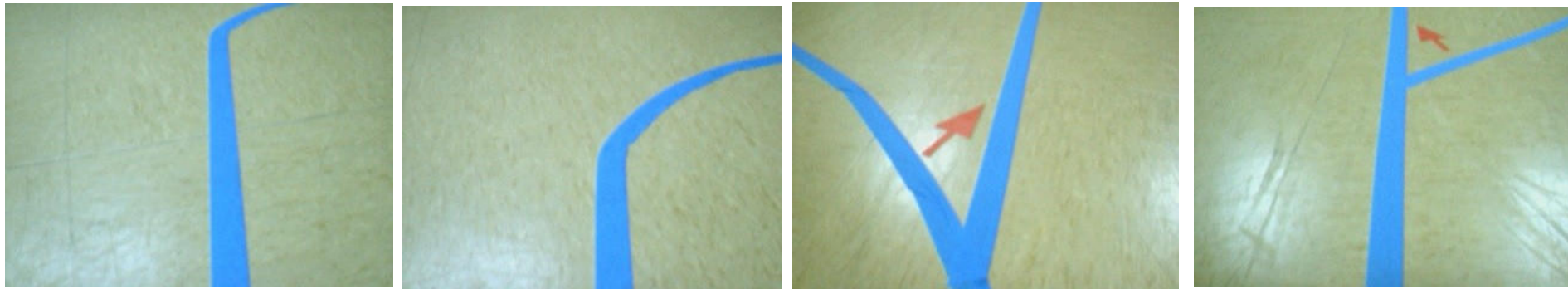
jamiesho@microsoft.com

shahrami@microsoft.com

Robótica y Percepción Computacional

Proyecto:

Control del robot para realizar el seguimiento de una línea y el encaminamiento a partir del reconocimiento de marcas.



Visión por Computador:

- Segmentación: línea, fondo, flecha y marca.
- Interpretación de la línea
- Interpretación de la flecha
- Extracción de información para calcular consigna control

Robótica y Percepción Computacional

Tareas para el próximo día:

Instalar paquetes relacionados con la asignatura (los robots tienen Ubuntu 24.04 LTS).

Puedes crear un entorno para las clases:

- `python3 -m venv clase`

E instalar los paquetes necesarios:

- `pip install numpy scipy matplotlib scikit-learn
opencv-python scikit-image jupyterlab imageio
pygame`

Tareas para la primera semana: Practica con funciones que lean una imagen y la muestren en una ventana.

OpenCV:

- `cv2.imread, cv2.imwrite, cv2.imshow,
cv2.waitKey(), cv2.destroyAllWindows()`
- `cam=cv2.VideoCapture(0), cam.set(...),
cam.read(...)`

Python:

- `scipy.misc.imread(), matplotlib.pyplot.imshow()`

Escribe un programa que lea continuamente una imagen de la cámara o de un archivo de vídeo y la muestre en una ventana.

Robótica y Percepción Computacional

Percepción computacional

1. Introducción (10 marzo)

Robótica, visión por computador,

2. Formación de imágenes digitales (7 abril)

Tecnología de formación de imagen, espacios de color

3. Segmentación de imágenes (14 abril)

Etiquetado de imágenes basado en color

4. Procesamiento y análisis imágenes (21 y 28 abril)

Filtrado, análisis de imagen,
extracción de información para control

5. Presentación de resultados y discusión (5 mayo)

Presentación en clase

Entrega análisis de imagen (11 mayo)

Robótica y Percepción Computacional

Percepción computacional

Evaluación (sobre un total de 25 puntos)

1. Entregas parciales (+ 5 puntos adicionales)
2. Entrega final (25 puntos)